A Novel and Efficient Surface Electromyography Decomposition Algorithm Using Local Spatial Information

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***Abstract*—Motor unit spike trains (MUSTs) decomposed from surface electromyography (sEMG) have been an emerging solution for neural interfacing, especially for the control of upper limb prosthetics. Accurate and efficient decomposition techniques are essential and desirable. However, most decomposition methods are designed for motor units (MUs) with global maximum of single or large muscle, while in general forearm muscles are usually small and slender with low global energy. Thus, we propose a novel approach using local spatial information towards more accurate and efficient sEMG decomposition of forearm muscles. A fast spatial spike detection method is proposed to replace the time-consuming iteration process of blind source separation (BSS) methods. Here, spatial distribution characteristics of motor unit action potential are leveraged to pre-classify the candidate MUs, and further to create initial MU templates, aiming to avoid repeating convergence to high-energy MUs. The results of both simulated and experimental sEMG signals show that low-energy MUs from small muscles are more easily found compared with conventional BSS algorithm. Specifically, the proposed method can identify more 40% reliable MUs while only 30% consuming time are needed. The outcomes provide a novel solution for more efficient sEMG decomposition, potentially paving the way of MUST-based non-invasive neural interface.**

***Index Terms*—Motor unit decomposition, multichannel surface EMG, spike detection.**

I. Introduction

TO ESTABLISH a reliable interface of prosthetic hands is one of the most difficult fields of human-machine interfacing (HMI), since hands are the most dexterous and complex parts of human body. Electromyography (EMG) signals have shown great potential for prosthetic hand control [1], [2], [3], for which contain large amounts of information of human body's motion intention. Myoelectric control schemes based on EMG have been widely developed to improve the dexterity of prosthetic hands [4], [5].

Muscles are directly innervated by motor neurons activation signals which is derived from the spinal cord [6], [7]. The neural drive to the muscle is the sum of the motor units action potential trains (MUAPTs) which are generated by spiking activities of motor neurons [8]. And the motor neurons discharge signal can be recorded by EMG electrodes as an instantaneous mixture of MUAPTs [9]. In the past decades, the MUAPTs can be automatically identified by the EMG decomposition techniques from EMG signals and used for myoelectric control system [10], [11], [12]. Farina et al. have shown that the control strategy based on EMG decomposition has better performance than classic myocontrol systems with higher intuitiveness and accuracy [10]. Moreover, EMG decomposition techniques have been successfully applied to precise prediction of finger and wrist movements [13], [14].

In order to decompose motor neurons discharges from intramuscular EMG (iEMG) or surface EMG (sEMG), variety of EMG decomposition methods have been developed by researchers. Generally, they can be classified into two categories: template-matching (TM) and blind source separation (BSS) method. Lefever et al. use TM method to identify individual MU firing from invasive bipolar needle electrode assuming that the signals generated by the same MU have similar shapes [15]. Nawab et al. improve the accuracy using integrated processing and understanding of signals (IPUS) framework [16], [17]. Further, the method is extended to five-pin sEMG sensor identifying motor units in a non-invasive way [18], [19]. TM methods have low computational complexity and can identify MUs online. However, since the performance of TM methods deteriorates with the increase of the number of electrodes [19], these methods can only identify MUs from fibers located close to several electrodes. On the other hand, development of multichannel sEMG with tens of electrodes make it possible to record large area muscle information synchronously. To deal with the large amount of signals, offline multichannel sEMG decomposition algorithms are proposed [20], [21], [22], [23]. Among these methods, a BSS method, called gradient convolution kernel compensation (gCKC) [21], is the most successful decomposition framework. They establish BSS model and use gradient descent

Manuscript received 10 April 2022; revised 28 July 2022 and 16 September 2022; accepted 23 September 2022. Date of publication 27 September 2022; date of current version 5 January 2023. This work was supported by the National Natural Science Foundation of China under Grants 52175021, 91948302, and 52205025. *(Corresponding author: Xinjun Sheng.)*

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Digital Object Identifier 10.1109/JBHI.2022.3210019

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method to iteratively identified individual MU. BSS-based methods have two major limitations, i.e., the repeated convergence to the same MU and the high computational complexity of iterative process. Some methods have been proposed to improve the decomposition performance by optimizing the convergence property of BSS-based methods. Chen et al. use a peel-off strategy to remove the estimated MUAPTs from raw signals and iteratively identify MUs from the residual signals [24]. Negro et al. project the estimated separation vector in the whitened orthogonal space [25]. Ning et al. use clustering method (K-means) to initialize innervation pulse trains (IPTs) [26], [27].

However, a significant negative correlation between the depth of muscles and the number of identified MUs by these methods and the decomposition ability decreases when MUs have less distant centroids [28]. It makes the performance of these methods degraded when applying to small and slender muscles with relatively low global energy. Moreover, for HMI implementations, when the results of BSS methods are used for analyzing data or pre-training the separation matrix of online decomposition approaches [29], [30], [31], the high computational complexity will increase the training time resulting in poor user experience. Therefore, there exists an urgent need for an accurate and fast decomposition method focusing on the complex muscle groups such as forearm.

In this study, we aimed to propose an efficient offline multichannel sEMG decomposition method which can identify more reliable MUs with low computational complexity. We proposed a spatial-based spike detection method using local information to pre-classify the spikes generated by different muscles to find more candidate MUs with relative low energy. Moreover, we used a low computational complexity density-based clustering algorithm to initialize separation vectors replacing the iterative process in BSS methods. Then we optimized the separation vectors and identified individual MU firings. The proposed method was validated by simulated and experimental data. And it was compared with gCKC which has been widely used for MU decomposition and chosen as benchmark by most multichannel sEMG online decomposition methods [29], [30], [31].

The remaining parts of this paper are organized as follows: Section II-A describes the data model of the algorithm. Sections II-B to II-E describe the algorithm details and framework. Section III describes the materials and methods used for validation. The simulated and experimental results and discussion are presented in Sections IV and V, respectively.

II. Theory and Algroithms

Fig. 1 shows the structure of the proposed local-information-based decomposition (LIBD) algorithm. The algorithm mainly comprises: spike detection based on spatial information, templates initialization by clustering spikes, templates optimization and MU firings identification. Firstly, the multichannel sEMG signal is filtered by band-pass filters (20–450 Hz). After decorrelating data by a zero-phase component analysis (ZCA) sphering method, called ZCA-cor whitening transformation, a sliding spatial window is employed to obtain sub-regions from electrode grid. Spatial spike detection is used to select spikes

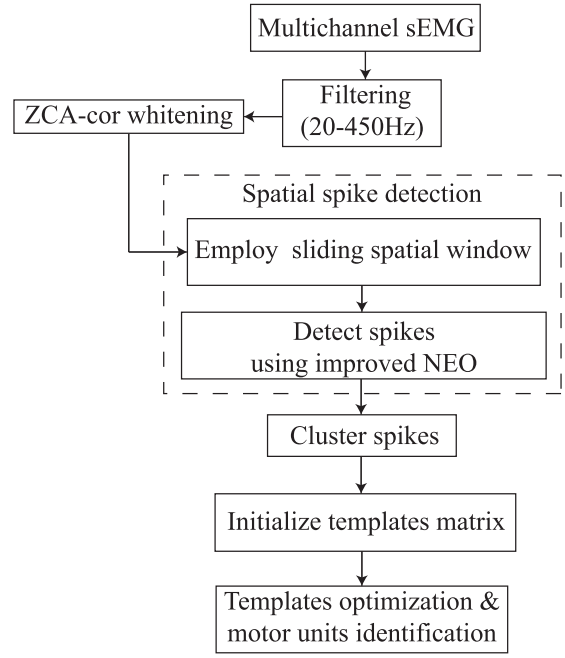

Fig. 1. The structure of the proposed multichannel sEMG decomposition algorithm.

with attenuation properties in all sub-regions. All spikes in different sub-regions are clustered respectively. Then, initialize templates by averaging spikes in each cluster. The initial templates are optimized by averaging segments that have peak correlation between templates and whitening data. Finally, the optimized templates are used to identify firings by calculating the correlation between templates and whitening data.

A. Data Model

Multichannel sEMG can be modelled as a linear time-invariant (LTI) system. And IPTs of MUs can be estimated by linear minimum mean square error (LMMSE) estimator [20]:

$$\hat{t_j}(n) = c_{t_j}^T C_{XX}^{-1} X(n) \tag{1}$$

where $C_{XX} = E[X(n)X^T(n)]$ is the covariance matrix and $X(n) = [x_1(n), x_1(n+1), \ldots, x_1(n+R-1), \ldots, x_M(n), \ldots, x_M(n+R-1)]^T$ describes an R delayed version of M channels containing the current and $R-1$ next samples at time n. c_{t_j} denotes the average waveform with R samples length of jth MUAP and can be calculated as follow [26]:

$$c_{t_j} = \frac{1}{card(s_j)} \Sigma X(s_j) \tag{2}$$

where s_j is firing times set of jth MU which is difficult to be obtained in advance. In order to estimate s_j fast and accurately, we proposed a detection method based on spatial information described in Section II-C.

B. Whitening

Similar to many BSS decomposition approaches [21], [25], raw EMG signals should be whitened before spike detection. We

used ZCA-cor Whitening, since it can maximally retain spatial information of spikes [32]. The whitened data can be calculated by:

$$Z = WX \tag{3}$$

where X is a matrix with $M \times R$ rows and N columns describing an R delayed version of M channels data with N samples and W is the whitening matrix. For ZCA-cor whitening, $W_{ZCA-cor}$ can be obtained as follow:

$$W_{ZCA-cor} = P^{-\frac{1}{2}} V^{-\frac{1}{2}} \tag{4}$$

where $V = diag(\sigma_1^2, \ldots, \sigma_{M\times R}^2)$ is variance matrix containing the variances $x_i x_i^T = \sigma_i^2$. The correlation matrix of X can be described as

$$P = V^{-\frac{1}{2}} C_{XX} V^{-\frac{1}{2}} \tag{5}$$

Substituting (3), (4) and (5) into (1) produces:

$$\hat{t_j}(n) = c_{t_j}^T (V^{\frac{1}{2}} P V^{\frac{1}{2}})^{-1} X(n) \tag{6}$$

$$= [c_{t_j}^T (P^{-\frac{1}{2}} V^{-\frac{1}{2}})^T][(P^{-\frac{1}{2}} V^{-\frac{1}{2}}) X(n)] \tag{7}$$

$$= (W_{ZCA-cor} c_{t_j})^T W_{ZCA-cor} X(n) \tag{8}$$

$$= w_{t_j}^T Z(n) \tag{9}$$

where w_{t_j} is the template vector of jth MU in the whitened space. And it can be obtained as:

$$w_{t_j} = \frac{1}{card(s_j)} \Sigma Z(s_j) \tag{10}$$

where s_j is firing times set of jth MU and Z is the whitened data.

C. Spatial Spike Detection

Spike detection is a technique to extract MU firing segments from noisy signals. Many spike detection methods were used for multi-channel sEMG decomposition. Ning et al. use an energy-based method [26]. Mohebian et al. and Gligorijević et al. extract segments whose maximum absolute values exceed a threshold [33], [34]. But all this methods have a tendency to miss low energy spikes when there were higher energy spikes discharging simultaneously.

Many of simultaneously firing MUs can be distinguished spatially from multichannel recordings. In order to detect different MU segments which discharged at the same time, we proposed an improved nonlinear energy operator (NEO) for multi-channel spike detection. The main idea is to pre-classify spikes to sub-regions with a 3×3 sliding spatial window based on the spatial distribution properties of MUs. The magnitude of an MU is dependent on the 'radial' position of the recording electrode and decreases fast from the nearest recording electrode to the electrodes perpendicular to the muscle fiber direction. Oeleveld et al. have shown that most energy of MU concentrates in a range of distance = 20 mm on the skin (perpendicular to the muscle fiber direction) [35]. In this study, we assumed that most energy of a spike concentrates in three electrode columns (interelectrode distance (IED) = 10 mm). On the other hand, since the MUAP conducts along fiber direction, there are phase differences among adjacent rows of channels [36]. It produces alignment error which will significantly reduce the accuracy of clustering. Therefore, we chose a compromise size of three rows. In each sub-region, the method only extracted spikes whose NEO value on the center electrode is higher than the values on the electrodes perpendicular to the muscle fiber direction. The mth electrode is the center electrode of the mth sub-region. Every electrode should be the center electrode of one of sub-regions detecting as many MUs as possible. Fig. 2(d) shows three different boundary conditions of sub-regions. The definition of original NEO [37] is given by:

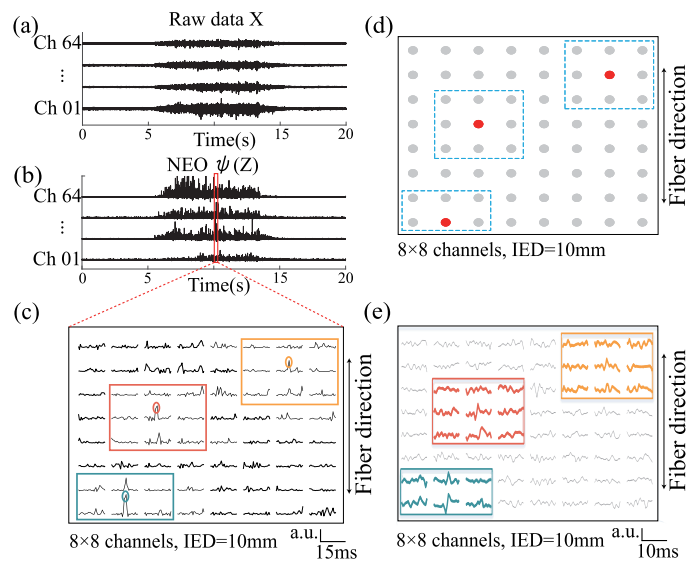

Fig. 2. Demonstration of spatial spike detection process. (a) 4 of 64 channels raw data recorded from experiment. (b) 4 of 64 channels NEO calculated by whitened data Z using (12). (c) Spatial spike detection of NEO in a 15 ms length samples segment. We assign a different color to each rectangle and ellipse to differentiate the three sub-regions. Ellipses mark the points of NEO that are selected to extract spikes. (d) Illustration of sub-region electrode sets with spatial window. A sub-region is depicted by a dotted rectangle and red points mark the center electrodes of different sub-regions. (e) Spikes extracted from whitened data Z using points selected by (c) with 20 samples (10 ms). Spikes with different color show the signal segments which are extracted from sub-regions of whitened data Z at points selected in (d). And the extracted spikes are pre-classified to the relevant sub-regions.

$$\psi\{x(n)\} = x^2(n) - x(n-1)x(n+1) \tag{11}$$

Assume that the whitened data $Z(n) = [z_{1_1}(n), z_{1_2}(n), \ldots, z_{1_R}(n), \ldots, z_{M_1}(n), \ldots, z_{M_R}(n)]^T$. Then, we defined the NEO of channel i as:

$$\psi_i\{Z(n)\} = z_{i_k}^2(n) - z_{i_{k-1}}(n) z_{i_{k+1}}(n) \tag{12}$$

where $k = round(\frac{R}{2})$ describes the middle point of R delayed samples on channel i. The detection threshold [38] of channel i was calculated by $thr_i = C \times \frac{1}{N} \Sigma_{n=1}^{N} \psi_i\{Z(n)\}$. If the value of $\psi_i\{Z(n)\}$ is higher than thr_i at time n_c, $Z(n_c)$ is a candidate spike with $M \times R$ samples representing R samples waveforms on M channels at time n_c. A spike was detected and classified to the ith sub-region, while fulfilling two conditions: 1) the value

of $\psi_i\{Z(n_c)\}$ was a peak higher than thr_i over R samples; 2) max absolute value of the ith channel segment of $Z(n_c)$ was higher than that of segments perpendicular to the channel i. An example of spatial spike detection is shown in Fig. 2.

D. Clustering and Templates Initialization

An improved version of density-based clustering method [39] was used to cluster detected spikes in different sub-regions respectively. Spikes activated by the same MU can be considered as a cluster consisting of points with similar shapes and small distance to each other in feature space. A cluster center has the highest local density in a dense area and a larger distance from all higher density points than their neighbors. Cluster number is automatically determined by finding all cluster centers. For each spike i, we calculated a decision score:

$$\gamma_i = \rho_i \delta_i \tag{13}$$

where ρ_i is the number of points whose distances to spike i are smaller than dc, and δ_i is the distance between spike i and the nearest point with higher density. We determined dc as follows. Two dimensional PCA features of spikes were used to calculate distances. Then, the distances were sorted in ascending order and represented by ds. The length of ds was Ns. dc was set equally to $ds(n)$ where $n = 0.5\% \times Ns$. Spike i and its neighbors were identified as cluster j, if γ_i was higher than a threshold $thr_{\gamma_i} = 4 \times median(|\gamma_i|)/0.6745$ [40]. s_{0j} was consist of the firing times of spikes in cluster j. And template w_{0t_j} was estimated by (10). An initial templates matrix $W_0 = [w_{0t_1}, w_{0t_2}, \ldots, w_{0t_{N_c}}]$ was obtained from all N_c clusters.

E. Templates Optimization and Motor Units Identification

The initial IPTs estimation of all N_c templates generated in Section II-D can be calculated by:

$$T_0 = W_0^T Z \tag{14}$$

where $T_0 = \left[\hat{t_1}^T, \hat{t_2}^T, \ldots, \hat{t_{Nc}}^T\right]^T$. W_0 was the matrix contained all N_c templates. Z was the ZCA-cor whitened data. The pulses set $s_j = \{n_{j_1}, \ldots, n_{j_u}\}$ of template w_j were estimated by the peaks of $\hat{t_j}$ which satisfied two conditions: 1) the peak value is higher than the threshold $thr_{\hat{t_j}}$ and 2) there's no higher peaks occurred within 10 ms. The threshold $thr_{\hat{t_j}}$ was determined as follows. The highest peak p_{t_j} of $\hat{t_j}$ was found. Candidate thresholds were calculated as $Candi_j = [0, 0.1p_{t_j}, \ldots, 0.9p_{t_j}]$. The average discharge rate and the coefficient of variation (CoV) for the interspike interval (ISI) were calculated by:

$$CoV_j = \frac{SD_j}{ISI_j} \cdot 100\% \tag{15}$$

where ISI_j is the mean ISI of the jth MUAPt estimated by t_j, and SD_j is the standard deviation of ISI [41].

The candidate threshold with minimum CoV_j was selected to be $thr_{\hat{t_j}}$. Then, the s_j was used to generated a new template w_{new_j} by using (10).

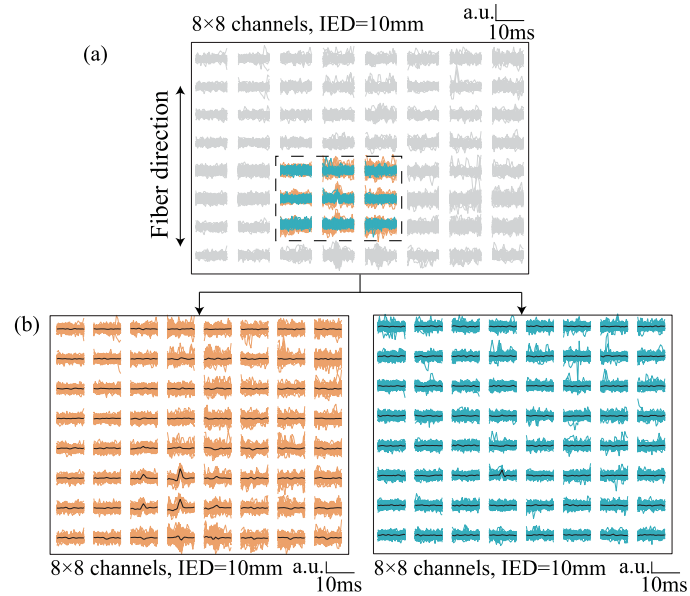

Fig. 3. An example of clustering and templates initialization process in one sub-region. (a) Spikes detected from the sub-region depicted by the dotted rectangle. Different color depicts different clustered spike groups. (b) The black solid lines represent the initial templates on each electrode generated by spikes in each group.

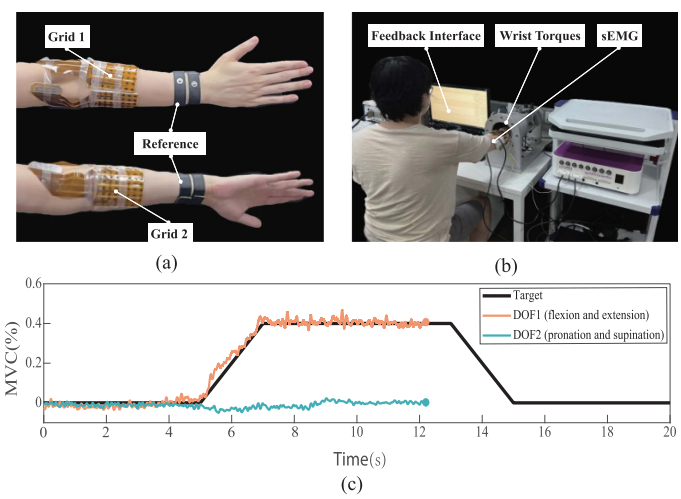

Fig. 4. Experimental setup. (a) The sEMG recording area. (b) Data acquisition platform. (c) Feedback interface to guide subjects with target and recorded torques in real time.

Optimal IPTs estimation T_h, templates W_h and pulses sets s_j of templates were obtained by repeating above steps for h times. Similar method was used in [26]. h in our algorithm was set to 3 which was much less than 20 in [26], due to better performance of the proposed spike detection and clustering method. Only the pulses with mean discharge rate 5-30 Hz [42] were retained. Pulses sets with Rate-of-Agreement (RoA) [43] >0.3 were merged. Then, IPTs estimation $T_{MU} = [\hat{t_1}^T, \hat{t_2}^T, \ldots, \hat{t_{N_{MU}}}^T]$, pulses sets $S_{MU} = \{s_1, \ldots, s_{N_{MU}}\}$ and templates matrix $W_{MU} = [w_{t_1}, \ldots, w_{t_{N_{MU}}}]$ of N_{MU} motor units were identified. Fig. 3 shows an example of clustering

Algorithm 1: The Proposed LIBD Algorithm.

Data: X with $M \times R$ rows and N columns (R delayed version of M channels sEMG and N samples)
Result: MU pulses sets $S_{MU} = \{s_1, \ldots, s_{N_{MU}}\}$

Compute Z by (3);
foreach *channel i of all M channels* **do** calculate NEO by (12) ;
Determine M sub-regions by sliding spatial window;
for $i \leftarrow 1$ **to** M **do**
  Detect candidate spikes from ith sub-region;
  Cluster spikes to obtain clusters $c_{i1}, c_{i2}, \ldots$;
end
Initial templates matrix $W_0 = [w_{0t_1}, w_{0t_2}, \ldots, w_{0t_{N_c}}]$ by all N_c clusters, $w_{0t_j} = \frac{1}{card(c_j)} \Sigma Z(c_j)$;
for $j \leftarrow 1$ **to** N_c **do**
  for $k \leftarrow 1$ **to** h **do**
    $w_{t_j} = w_{0t_j}$;
    $t_j = {w_{t_j}}^T Z$;
    Estimate pulses set s_j;
    Update $w_{t_j} = \frac{1}{card(s_j)} \Sigma Z(s_j)$;
  end
  if $5 < MDR < 30$ && $CoV < 0.4$ **then**
    Accept the jth source estimation;
  end
end
Merger pulses sets with $RoA > 0.3$;

and templates initialization following spatial spike detection with a 3×3 spatial window.

The pseudocode of the proposed decomposition method is shown in Algorithm 1.

III. Materials and Methods

A. Simulated Data

Simulated data were generated by a multi-layer cylindrical volume conductor model [44] in a grid of 8×8 monopolar surface electrodes with 10 mm IED in both directions. Three excitation levels (10%, 20%, 30%) were conducted 5 runs with different white Gaussian noise levels (20 dB, 25 dB, 30 dB, without noise) respectively. The input excitation was constant and lasted for 20 s. All these 60 sets data (3 excitation levels $\times$ 4 noise levels $\times$ 5 runs) were sampled at 2048 Hz.

The identified MUs with the RoA >30% were used to evaluate the decomposition accuracy of simulated data by calculating the sensitivity and the precision [45]:

$$Sensitivity = \frac{TP}{TP + FN} \tag{16}$$

$$Precision = \frac{TP}{TP + FP} \tag{17}$$

where TP is the number of matched discharges between decomposition results and the simulated firings. FN denotes the number of missed discharges, and FP is the number of incorrectly identified discharges. Another signal-based performance metric, the pulse-to-noise ratio (PNR) [45], was also calculated for every identified MUs by:

$$PNR(j) = 10log\left(\frac{E({\hat{t_j}}^2|_{\hat{t_j} \geq r})}{E({\hat{t_j}}^2|_{\hat{t_j} < r})}\right) \tag{18}$$

where $E(\hat{t_j}|_{\hat{t_j} \geq r})$ and $E(\hat{t_j}|_{\hat{t_j} < r})$ denote the mean across all time moments in which the jth MU is estimated to have or not to have discharged, respectively. The function $\hat{t_j}$ is computed by (9). A regression line $y = a \cdot log(b(x + c)) + 100$ was fitted with the nonlinear least-squares fitting method to show the relation between the sensitivity/precision (y) and PNR (x), where a, b and c were the parameters to be estimated.

B. Experimental Data

Seven able-bodied subjects (six male, one female, aged 25 ± 3) participated in the experiment. All of them were recruited from Shanghai Jiao Tong University and were presented with an informed consent form. The experimental protocols abided by the Declaration of Helsinki and were approved by the ethics committee of Shanghai Jiao Tong University (approval number: B2020026I).

The experimental data were acquired from the dorsal and the volar of the forearm as depicted in Fig. 4(a). Two electrode grids of 64 channels (8×8, ELSCH064NM3, OT Bioelettronica, Italy) were used to record surface EMG signals. Interelectrode distance and electrode diameter were 10 mm and 4 mm, respectively. The wrist torques were synchronously acquired by a home-made torque measurement device. The data acquisition platform was shown in Fig. 4(b).

Subjects accomplished three trials for each four wrist isometric contractions (extension, flexion, pronation and supination) following a trapezoid force trace. For each trial, the subject was instructed to perform isometric contraction under the guidance of the feedback interface shown in Fig. 4(c). The excitation level increased from 0 to 40% of the maximum voluntary contraction (MVC) in 2 s, remained constant for 6 s, and decreased to 0 in 2 s. The data of each grid were decomposed separately. The average number of MUs identified by both methods were counted independently for different wrist motions. The MDR and CoV for ISI were analyzed for each motion. Any point with unusually ISI (>200 ms or <20 ms) was removed as outliers when MDR and CoV were calculated. Only MUs with $CoV \leq 40\%$ were considered since MUs with $CoV > 40\%$ are typically associated with unreliable results [46]. Differences in MDR and CoV of MUs identified by different decomposition methods were tested by a one-way ANOVA. Continuous variables were shown as mean Â± std. The level of significance was set to $p = 0.05$. PNR was also calculated for all MUs as a metric evaluating the decomposition results.

To evaluate the decomposition quality, the signal-to-interference ratio (SIR) [41] of each channel was calculated as

follow:

$$SIR(i) = \left(1 - \frac{E\left[(x_i(n) - \sum_j z_{ij}(n))\right]^2}{E[x_i^2]}\right) \cdot 100\% \quad (19)$$

where $x_i(n)$ was the signal from ith channel, $\sum_j z_{ij}$ was the signal reconstructed by all identified MUAPTs.

C. Decomposition Time

The decomposition time of both methods was recorded respectively as a measure of algorithm complexity and efficiency. The simulated data and the experimental sEMG data were decomposed by gCKC method and the proposed method, respectively. The number of gCKC first iterations for searching different MUs was set to $N_m = 30$. And the number of gradient descent iteration was set to $N_{g1} = 25$ before $h_g = 5$ times separation vector optimization by (2). Then $N_{g2} = 5$ times more gradient descent iteration was performed. Most of these parameters are empirical, and come from forearm sEMG decomposition experience. The number of templates optimization iteration in proposed method was set to $h = 3$. To reach the best accuracy for the BSS-based methods, the extension factor was set to 20 for both methods [25]. Both methods were programmed in Matlab (R2021b). Vectorized code was used as possible to avoid looping for better performance. The hardware used for decomposition was a PC with Intel Core i7-8700 CPU and 16 GB of RAM.

IV. Results

A. Simulation Results

The results for simulated data decomposed by the proposed method and gCKC are shown in Figs. 5 and 6. For simulated data with excitation level of 10%, 20%, 30% under none added noise condition, the identified MUs using the proposed method were 22.0±1.9, 25.4±1.9, and 26.0±1.2, with sensitivity >95% and precision >98%, while those of gCKC were 17.8±1.3, 18.8±1.5, and 18.6±1.9, respectively. The number of identified MUs decreased to 14.0±0.5 (12.4±1.1, 9.8±1.3) and 11.6±1.3 (9.2±0.4, 7.8±1.3) when the SNR was decreased to 30 dB (25 dB, 20 dB). Both methods provided similar sensitivity and precision at excitation level of 10% and 20%. As excitation level increased to 30%, the proposed method provided about 5% higher sensitivity and precision. And in all simulated conditions, the proposed method identified more MUs correctly.

For all excitation levels, the proposed method cost 18 s in average while gCKC cost about 77 s shown in Fig. 6(a). Fig. 6(b) illustrates the relation between the decomposition time and the number of correctly identified MUs by both methods. The time used by the proposed method was significantly less than gCKC when the same number of MUs were correctly decomposed.

The sensitivity and precision of all excitation and SNR levels are shown in Fig. 7, as a function of PNR. It demonstrated that the MUs with PNR >28 dB corresponded to high sensitivity and precision. Similar results were also reported in [45] and [29].

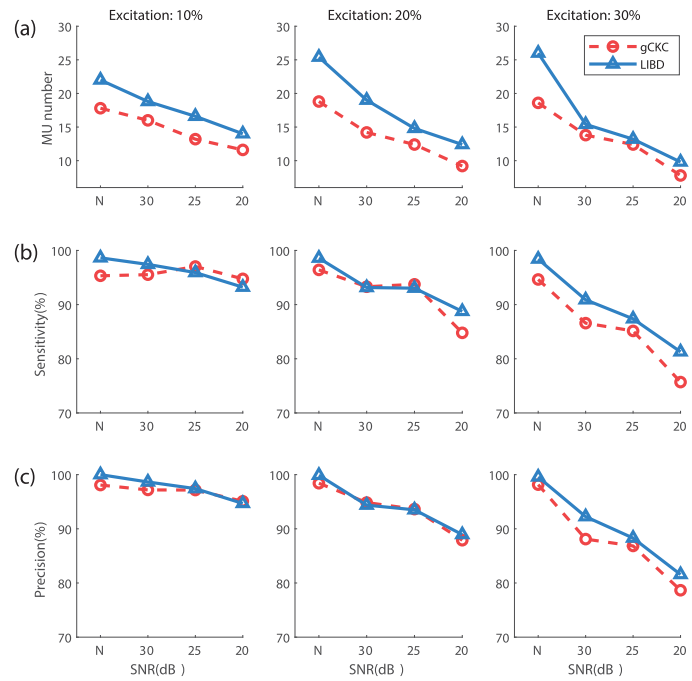

Fig. 5. Decomposition results for the simulated data. (a) The average number of correctly identified MUs under different excitation and noise level. (b) and (c) illustrate the average sensitivity and precision of the identified MUs. N in the horizontal axis denotes conditions without noise.

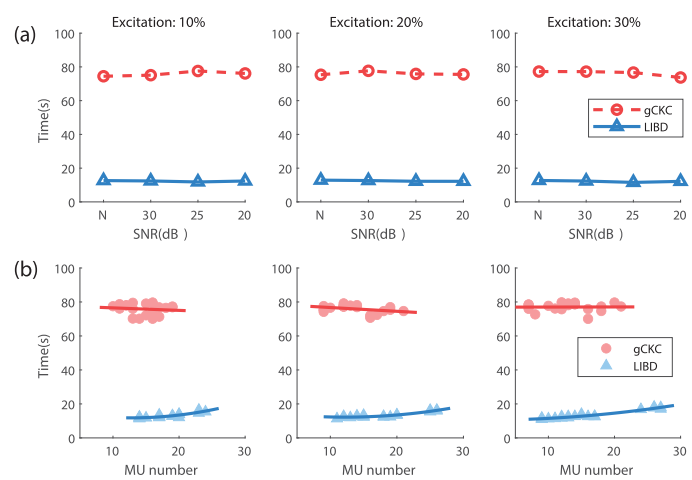

Fig. 6. Decomposition time for the simulated data. (a) The average decomposition times of gCKC and the proposed method under different conditions. (b) Decomposition times against identified MU number, and the fitting curves of gCKC and the proposed method.

B. Experimental Results

Table I summarizes the average number of identified MUs with $CoV < 40\%$ and decomposition time. For all four wrist tasks, the proposed method identified more 3-4 MUs in average but cost less than one third of the time used by gCKC. Fig. 8 depicts an example of MU discharge patterns identified by both methods from wrist extension with 40% MVC. Vertical lines represent MU discharge instants. Blue lines indicate MUs identified by our method while red lines represent the results

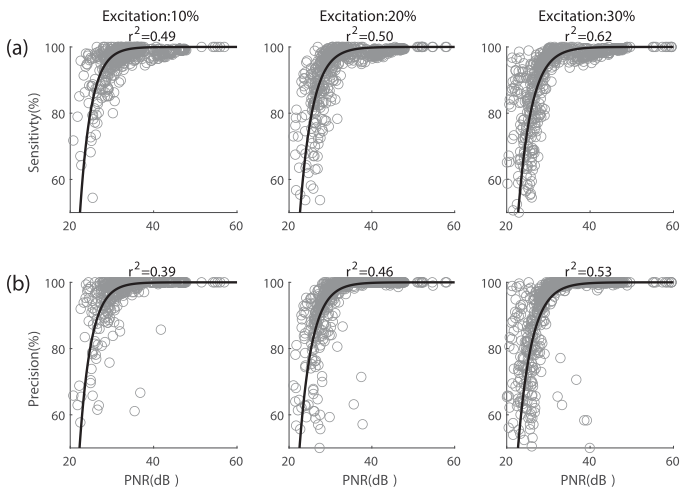

Fig. 7. Relation between the PNR and two decomposition accuracy metrics. All simulation results with different excitation levels are accumulated.

TABLE I
THE NUMBER OF IDENTIFIED MOTOR UNITS AND DECOMPOSITION TIME

	No.MUs		Time (s)	
Task	gCKC	LIBD	gCKC	LIBD
Flexion	10.1±5.1	14.1±8.5	127.4±16.0	46.5±13.9
Extension	6.6 ±4.5	9.8 ±6.3	130.3±19.4	36.9±8.2
Pronation	6.2 ±3.5	10.7±5.2	129.8±16.4	35.5±11.0
Supination	7.9 ±4.8	10.4±6.1	122.6±11.6	40.1±10.3

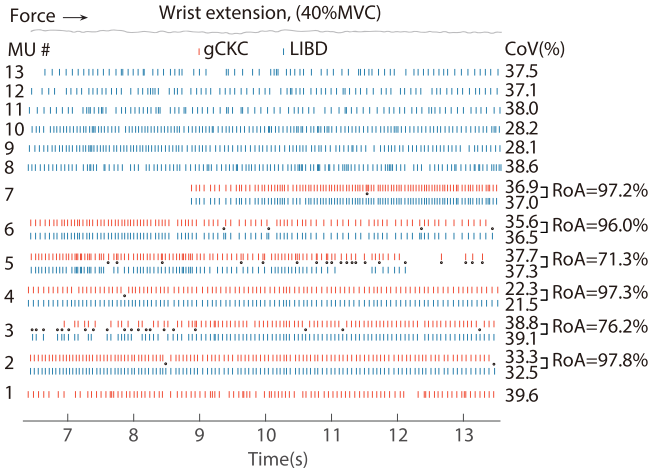

Fig. 8. A comparison of discharge instants of the proposed method and gCKC method. Blue lines indicate MUs identified by the proposed method while red lines represent the results decomposed by gCKC.

decomposed by gCKC. SIR and PNR, as defined in (19) and (18), are shown in Table II. The average PNR of MUs identified by both methods were higher than 30 dB which usually correspond to high decomposition accuracy and sensitivity (the relation was investigated in Fig. 7). The SIR reported in Table II are the mean values of all 64×2 channels. In fact, the SIR estimated the percentage of the energy of the single-differential data reconstructed

TABLE II
DECOMPOSITION ACCURACY METRICS

	SIR (%)		PNR (dB)	
Task	gCKC	LIBD	gCKC	LIBD
Flexion	13.7±7.5	16.6±8.5	31.6±3.7	31.0±3.1
Extension	10.3±5.3	11.5±7.2	32.6±3.1	31.6±2.8
Pronation	8.3 ±4.0	11.6±4.5	32.7±3.7	31.4±3.5
Supination	12.6±6.4	15.3±9.6	32.0±2.8	31.0±2.2

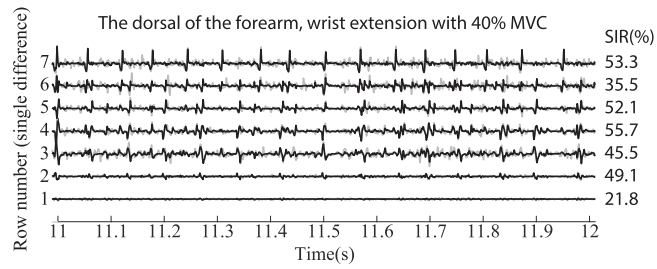

Fig. 9. The single-differential sEMG from a central column electrodes on the dorsal of the forearm (gray) and the sum of MUAP trains identified by the proposed decomposition method (black). The SIR of the 7 sEMG channels are shown ranged between 21% and 56%. The averaged SIR across all the channels is 18%, ranged between 3% and 80%.

TABLE III
DISCHARGE CHARACTERISTICS OF IDENTIFIED MOTOR UNITS

	CoV (%)		MDR (pps)	
Task	gCKC	LIBD	gCKC	LIBD
Flexion	28.9±5.9	30.4±6.0	12.9±3.0	13.1±2.6
Extension	29.2±6.5	30.5±6.4	14.5±4.2	14.5±4.3
Pronation	29.8±6.1	31.8±5.9	15.8±4.8	15.8±5.1
Supination	27.2±6.8	28.4±7.0	13.2±3.1	13.7±3.6

by decomposition results. Our method provided slightly higher SIR since more MUs were identified.

Fig. 9 shows an example of experimental sEMG detected by a central column electrodes (gray) and the reconstructed signals by sum of MUAPTs (black). The MDR and CoV for ISI of the MUs are summarized in Table III. These two characteristics from gCKC and the proposed method were not significantly different for all tasks ($p > 0.05$). It means that the neural discharge information are estimated equivalently by both methods.

V. DISCUSSION

This study proposed a novel and efficient MU decomposition method for multichannel sEMG. Simulated and experimental signals were used to validate and compare the results decomposed by BSS method and the proposed LIBD. Fast ICA [25] and gCKC [21] are the most representative and widely used BSS methods. When both methods reach their best performance, fast ICA identified slightly more MUs with significantly higher time complexity than gCKC because of different iterations (200 for fast ICA [47], 30 for gCKC). For fairly comparing both of the

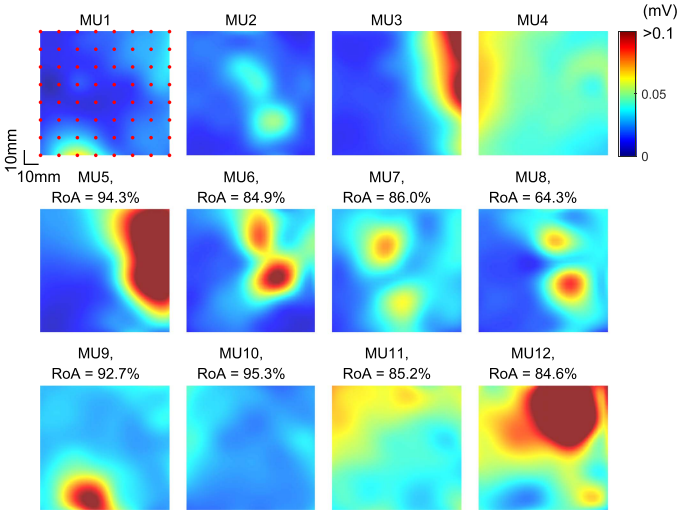

Fig. 10. MUAP peak-to-peak amplitude map. 12 MUs decomposed from experimental data of 8×8 electrodes are shown. MU1-MU4 were identified only by the proposed method. MU5-MU12 were identified by both methods.

decomposition ability and time complexity with BSS method, gCKC is chosen as baseline algorithm. Most decomposition methods of multichannel sEMG [21], [24], [25], [27] can be unified into one framework containing two main parts: (a) estimating initial c_{t_j} of MU_j in (1); (b) refining c_{t_j} by (2) or other methods and identifying MUAPTs from $\hat{t_j}$ in (1). In isometric contractions, the accuracy of decomposition results are mainly affected by the MUAPs (c_{t_j} in (1)) which are assumed constant and can be accurately obtained by (2) in part (b). Therefore, the same strategy described in Section II-E are used to refine c_{t_j} and identify MUAPTs from $\hat{t_j}$ for both methods. The results demonstrate that the proposed method can identify more reliable MUs and cost less time under isometric contractions with different conditions.

A. Number of Reliable Motor Units

A major problem of classical BSS approaches is that the algorithms always converge to the same source repeatedly and favor the MUs with high energy. Several methods have been proposed to solve this limitation. Chen et al. used a peel-off strategy iteratively removing the estimated MUAPts from the raw data [24]. But the peel-off process increases the noise level of residual signals rapidly and reduces decomposition accuracy. Another effective but computational complexity method proposed by Negro et al. [25], introduces an orthogonalization step for the iteration of estimating new sources. It showed advantages for avoiding converging to same sources, but the orthogonalization method also has a weak ability to identify low energy sources due to the energy-based initialization method. Ning et al. and Mohebian et al. use clustering methods to determine the initial estimation of the MUs [26], [33]. Clustering-based methods have better convergence properties, but more iterations (30 in clustering method [26], about 5 in iterative estimation method [25]) are needed to refine the source estimation because of lower accuracy of initial points. These methods tend to identify MUs with high values of summation of all channels' energy. It works well for muscles such as FDI or gastrocnemius, but the performance usually degrades for complex condition such as forearm muscles which are usually small and slender with lots of low global energy MUs. To deal with this problem, we proposed a spatial spike detection method using local information which is more sensitive to MUs with low global energy from multichannel sEMG. Algorithms described in Sections II-B to II-D focus on the local maximum using local information to find more MUs recruited by small muscles. Fig. 5(a) and Table I show that the proposed method can identify more MUs in different conditions of simulated data and experimental data. MUs identified by both methods have similar reliability shown in Fig. 5(b)–(c), Tables II and III. Fig. 10 shows an example of the peak-to-peak value maps of MUs decomposed from experimental data. MU1-MU4 are identified only by the proposed method, and MU5-MU12 are identified by both methods. In Fig. 10, MU1-MU4 and MU10-MU11 have relative low values of summation of all channels' energy, but only MU10-MU11 are identified by both methods. It shows the ability of our proposed method to find more MUs with relative low energy.

B. Decomposition Time

The decomposition model (9) was derived from the LMMSE estimator in (1). The proposed method used a clustering-based method to directly estimate initial w_{t_j} in (9) with low complexity instead of estimating c_{t_j}. Such method is usually used by template-matching methods to decompose sEMG signal with less channels (usually less than 5). We proposed an improved scheme that allows the clustering-based method to be applied to multichannel sEMG signals(64 channels or more). It can get all candidate w_{t_j} by clustering spikes avoiding high complexity iterative process.

The proposed method mainly consists of four parts presented in Sections II-B to II-E. And the time complexity of the proposed method mainly contributed by templates optimization proposed in Section II-E. The first two parts generate an MR-by-N_c matrix describing N_c initial templates for R delayed version of M channels signals. For 64 channels forearm sEMG signals, N_c usually ranges from 100 to 200. Templates optimization is an iteration process optimizing the initial templates matrix and identifying MU firings. With (14), we multiply the N_c-by-MR matrix ${W_0}^T$ and the MR-by-N matrix Z by h times with $O(N_c MRNh)$ time complexity. For gCKC method, the mainly iteration process performs the matrix multiplication depicted in (1) by $N_{g1} + N_{g2} + h_g$ times with $O(MRN(N_{g1} + N_{g2} + h_g))$ for each of N_m candidate MUs. Thus, the entire gradient descent iteration process costs $O(N_m MRN(N_{g1} + N_{g2} + h_g))$. For the parameters given in Section III-C, time complexity of iteration process is reduced from $O(1050MRN)$ to $O((300 \sim 600)MRN)$. The comparisons of decomposition time in Fig. 6 and Table I also show that the decomposition time cost by the proposed method is significantly less than the time used by classical BSS methods for simulated and experimental data.

C. Applications and Limitations

Our method is used to decompose MUs from multichannel sEMG signals. The method can also be used to establish the EMG-based HMI, especially for prosthetic hands because of the significant advantage in efficiency (more MUs and less decomposition time) for the sEMG signals with lots of small muscles. And the decomposition results can be used to train online decomposition methods [29], [30], [31]. Since the proposed method identifies more reliable MUs, the performance of these online methods can be improved. In this paper, since the proposed method is designed for forearm muscles, we didn't evaluate the ability for large muscles. Moreover, the MUs are also assumed to be constant like all other BSS methods, therefore the decomposition ability of the proposed method decreases for dynamic non-isometric contractions. Future work will seek to design an update strategy for inconstant MUs to address this limitation.

VI. Conclusion

We developed a novel and efficient offline decomposition method for multichannel sEMG signals. A spatial-based spike detection and templates initial method was proposed to replace the iteration process of classical iterative BSS method. The results indicated that the proposed method to decomposed more reliable MUs by avoiding the limitations that the classical methods have a high computational complexity and always converge to the same source with high energy. Since the process of generating templates is simplified, the proposed decomposition method costs only one-third of the time used by classical BSS method under several conditions.

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